OPEN SOURCE HF SDR TRANSCEIVER

mcHF v0.9

User Manual

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Covers application firmware v0.54.93 (M7) and baseband firmware v2.12.9 (M4)

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Hardware and firmware are released under the GNU General Public License v3.

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1 About this manual

The mcHF v0.9 is a work in progress. The firmware changes every week, and some features described here are new. This manual describes the radio as it is now, not as it will be. Each feature has a status mark:

WORKING	Tested on the radio and in regular use.
BETA	Implemented and partly tested. Expect rough edges, and please report what you find.
PLANNED	Visible in the user interface, or planned, but not finished. Do not rely on it yet.

The screen pictures were taken on the radio itself with the built-in screenshot function (see Screenshots), so they show exactly what the display shows.

> **You need a licence to transmit**
>
> Only transmit if you hold an amateur radio licence, and only on frequencies, modes and power levels your licence allows. Receiving needs no licence.

2 The radio at a glance

The mcHF is a self-contained software defined radio (SDR) transceiver for the HF amateur bands. It is designed for portable and off-grid use. The battery, the display and all the signal processing are inside the case, so it needs no computer, mobile network or internet connection.



The mcHF v0.9

2.1 Main specifications

Frequency coverage	Amateur bands 160 m to 10 m (1.8 to 30 MHz)
Modes	SSB (LSB/USB), CW, AM and synchronous AM; FM in the DSP (no front-panel selection yet); FT8; MarsChat
Output power	Up to 25 W PEP, in steps from 0.5 W to 20 W
Display	4.3" IPS LCD, 800 × 480 pixels, capacitive touch
Power	Internal 80 Wh lithium battery (5 cells in series), with a smart battery gauge and a USB-C PD charger
Processor	STM32H747 dual core: a Cortex-M7 at 480 MHz runs the user interface, and a Cortex-M4 runs the DSP
DSP	Based on the UHSDR firmware: WDSP AGC, noise reduction, noise blanker, automatic and manual notch, speech compressor, iambic CW keyer

Auxiliary radio	LoRa sub-GHz transceiver, used for MeshCore text messaging
Time	Battery-backed real-time clock. On rev B boards, a GPS receiver sets the clock.
Storage	microSD card for logs, settings backups and screenshots
Other	Bluetooth audio (in development), two USB-C ports

2.2 Two radios in one box

The mcHF contains two independent radios:

- **The HF radio** is the main transceiver. It is shared by the normal radio screen and the digital-mode screens (FT8, MarsChat). Only one of these screens controls the HF radio at a time.
- **The LoRa radio** is a separate sub-GHz transceiver. MeshCore runs on it and can keep receiving in the background while you use the HF radio. Messages that arrive appear as notifications on the spectrum display.

3 Controls

3.1 Front-panel keys

The front panel has these keys:

- **Top left:** POWER, BAND− (previous band) and BAND+ (next band)
- **Bottom left:** the volume encoder, then MODE and DSP
- **Below the display:** F1 to F5, evenly spaced

A short **press** does the main function. Holding a key for more than 0.4 s is a **hold**, which is a separate action and does not also do the press action.

Key	Press	Hold
POWER	Switches the radio on and off. See Power on and off.	—
BAND−	Previous (lower) band	—
BAND+	Next (higher) band	—
MODE	Opens the quick-access panel (band, mode, step, filter). Press again to close it.	—
DSP	Opens the Audio panel. Press again for the AGC / Attenuator panel, and again to close it.	—
F1	Opens or closes the Menu	Saves a screenshot to the SD card
F2	Quick launch: opens or closes the app set for it. Default: FT8	—
F3	Quick launch. Default: MeshCore	—
F4	Quick launch. Default: MarsChat	—
F5	**TUNE:** switches the transmitter on and off with a steady carrier. See Tune.	—

> **F5 keys the transmitter**
>
> One press of F5 starts transmitting a carrier, and it keeps transmitting until you press F5 again. Connect an antenna or a dummy load before you press it.

Quick launch keys. F2, F3 and F4 can each open any app: FT8, MeshCore, MarsChat or WSPR, or nothing (None). Choose them in the User Interface menu. A key opens its app from the main screen, and the same key closes it again. While an app is open, the keys of the other apps do nothing. Close the app first, then open the next one. WORKING

3.2 Rotary encoders

The encoders have no push function.

Encoder	Function
Tuning dial	Tunes the active VFO by the current tuning step. Each band remembers its own step. **Shake to change the step:** with the step at 1 kHz or 500 Hz, turn the dial quickly back and forth (about four changes of direction in under a second). The step toggles between 1 kHz and 500 Hz, and the frequency stays where it was before the shake. With a finer step, rocking the dial around a station does nothing special. WORKING
Volume encoder	Audio volume on receive (0–20). In CW transmit it sets the sidetone level instead. In some dialogs it moves the selected slider.

3.3 Touch screen

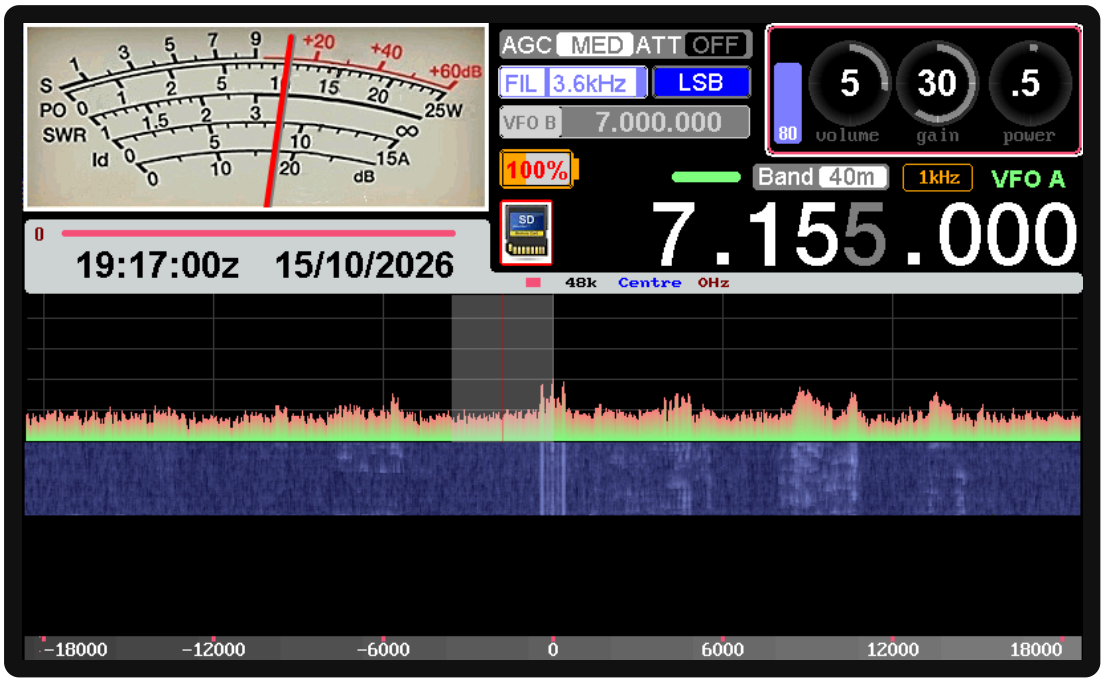
All dialogs and menus work by touch. On the main screen, touching the band guide above the spectrum jumps to that part of the band (for example the CW, digital or phone segment).

3.4 Power on and off

Press the **POWER** key in the top left corner to switch the radio on. To switch it off, press POWER again, or use **Shutdown** on the BMS tab of the Battery Manager. The radio saves its settings (band, frequencies, mode and so on) in battery-backed memory a few seconds after you change them, and writes them to the settings file on the SD card when you switch off. BETA

4 The main screen

After power on the radio shows the main screen. You operate the HF radio from here.



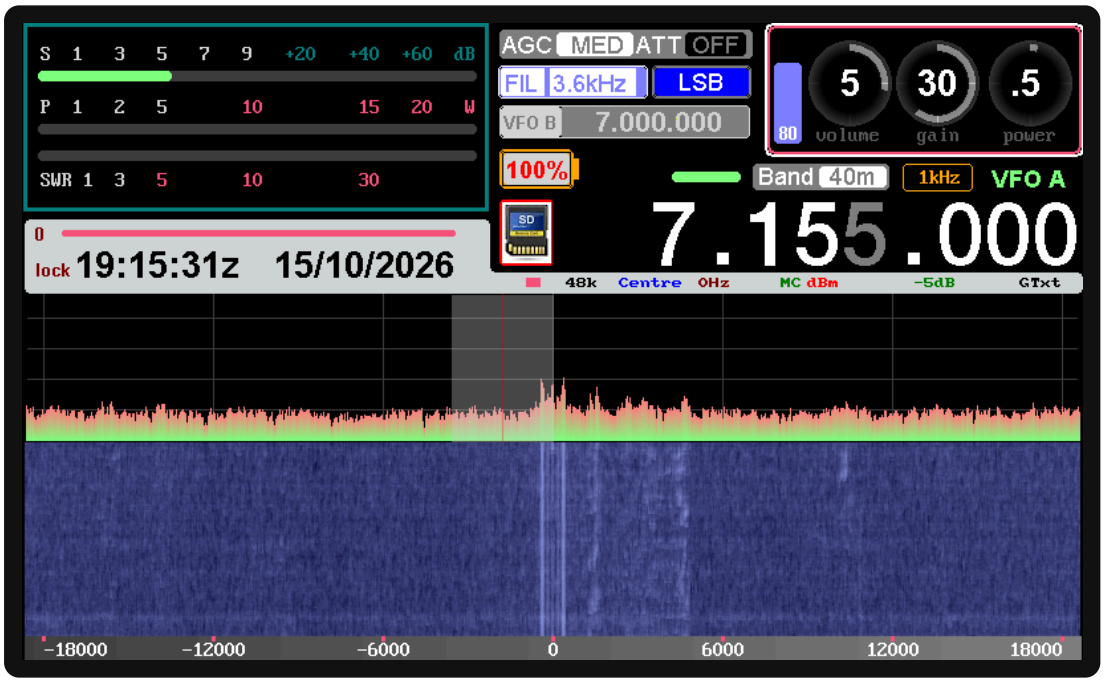
The main screen with the analogue S-meter, receiving on 40 m

4.1 Top row

Element	What it shows	Status
S-meter (top left)	Analogue needle meter with realistic ballistics, or a bar meter with S, power (P) and SWR scales. It measures only the signal inside the receive passband. Choose the style in User Interface.	BETA not yet calibrated: it reads high
Clock	UTC time and date. *lock* appears when GPS has set the clock.	WORKING
AGC / ATT	The AGC mode (SLOW, MED, FAST, CUSTOM, OFF) and the attenuator setting	WORKING
FIL / mode	The receive filter bandwidth and the demodulation mode (LSB, USB, CW, AM)	WORKING
Second VFO	The frequency of the VFO that is not active	WORKING
Battery	Battery charge in percent	WORKING
SD card icon	Shown when a card is inserted and mounted	WORKING
Knobs (top right)	**volume** (audio volume), **gain** (RF gain) and **power** (transmit power in watts). The small bar on the left shows the processor load.	WORKING

4.2 Frequency and spectrum

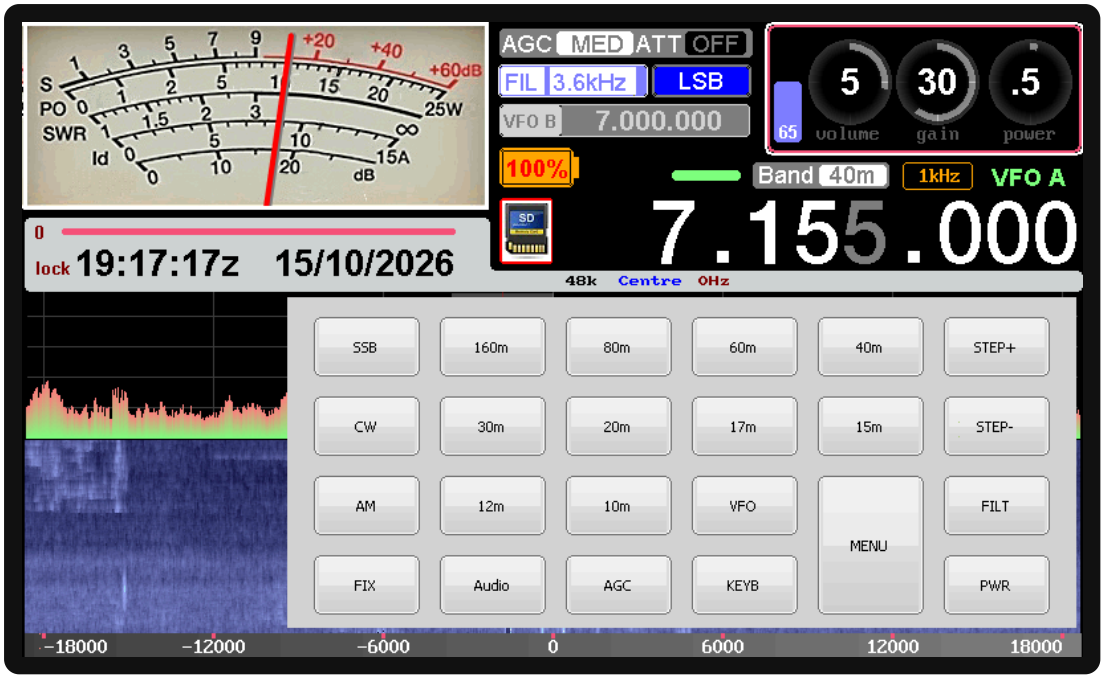
Element	What it shows	Status
Band / step / VFO	The current band, the tuning step and the active VFO (A or B)	WORKING
Frequency	The receive frequency of the active VFO. The digit that the tuning step changes is highlighted.	WORKING
Scope info line	The displayed span (48 kHz), the spectrum mode (Centre or Fixed) and the tuning offset	WORKING
Spectrum scope	Real-time spectrum around the receive frequency. The grey band shows the receive passband, and the red line marks the receive frequency.	WORKING
Waterfall	Spectrum history under the scope	WORKING
Frequency scale	Offset from the receive frequency in Hz, ±24 kHz	WORKING
Notifications	Messages fade in over the spectrum: MeshCore messages, saved screenshots	WORKING



The main screen with the bar S-meter

4.3 The quick-access panel (MODE key)

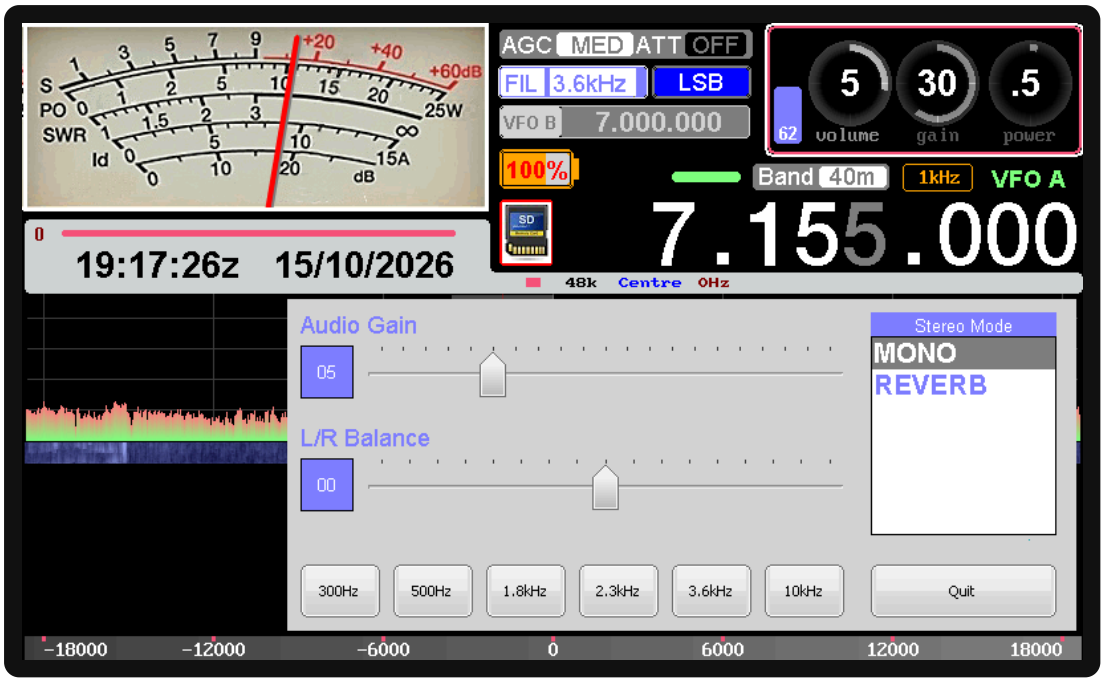
The **MODE** key opens a touch panel with the most-used controls. Touch a button to use it. Press MODE again to close the panel.



The quick-access panel

Button	Function
SSB	SSB, with the usual sideband for the band: LSB below 10 MHz, USB above
CW	CW mode
AM	AM mode. If *AM as SyncAM* is on in the Baseband menu, AM uses the synchronous detector.
160m … 10m	Go straight to a band. Every band remembers its own frequency, mode, filter, volume, step and power.
STEP+ / STEP−	Tuning step, from 1 Hz to 10 MHz in factors of ten
VFO	Swap between VFO A and VFO B
FILT	Step through the receive filters (300 Hz to 10 kHz)
FIX	Switch the spectrum between *Centre* mode (the receive frequency stays in the middle and the spectrum scrolls) and *Fixed* mode (the spectrum stands still and the receive marker moves across it)
Audio / AGC	Open the Audio or AGC / Attenuator panel
KEYB	Reserved PLANNED
MENU	Open the Menu
PWR	Step through the transmit power levels: 0.5, 1, 2, 5, 10, 15 and 20 W

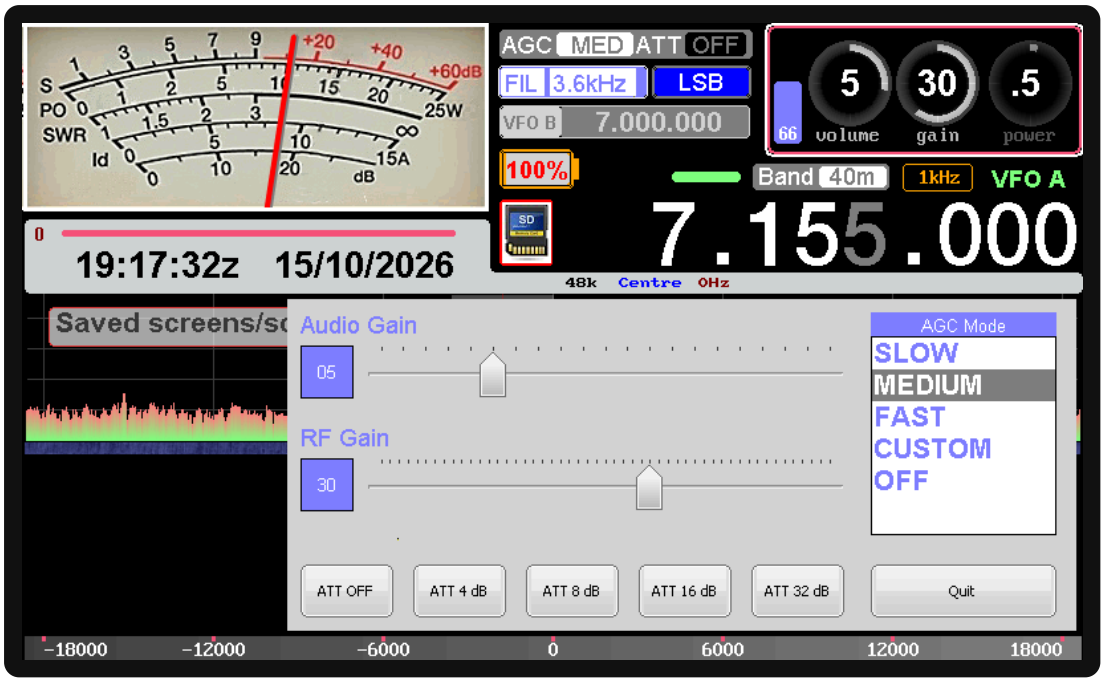
4.4 Audio panel (DSP key)



The Audio panel

- **Audio Gain**: the receive audio level, the same as the volume encoder
- **L/R Balance**: the headphone balance
- **Filter buttons** (300 Hz, 500 Hz, 1.8 kHz, 2.3 kHz, 3.6 kHz, 10 kHz): select the receive bandwidth directly. Use 300 Hz or 500 Hz for CW, 1.8 to 2.3 kHz for SSB, and 3.6 to 10 kHz for AM.
- **Stereo Mode**: MONO, or REVERB for a spacious effect on headphones
- **Quit** closes the panel. The DSP key does the same.

4.5 AGC / Attenuator panel (DSP key, second press)



The AGC / Attenuator panel

- **Audio Gain**: the receive audio level, as on the Audio panel
- **RF Gain**: sets the AGC threshold, the signal level where the AGC starts to act. Lower it to quieten band noise on strong-signal bands. With AGC off, it sets a fixed receiver gain.
- **AGC Mode**: SLOW (with 1 s hang), MEDIUM, FAST, CUSTOM (long, 2 s) or OFF. The decay times can be changed in the Baseband menu.
- **ATT**: input attenuator, OFF, 4, 8, 16 or 32 dB, for very strong signals
- **Quit** closes the panel

BETA The AGC and RF gain controls were reworked for the new DSP and still need more listening tests.

5 Receiving

1. Connect an antenna.
2. Select a band with BAND+ / BAND−, or with the band buttons on the quick-access panel.
3. Select the mode (SSB, CW or AM) on the quick-access panel.
4. Tune with the tuning dial. Change the tuning step with STEP+ / STEP−. 100 Hz or 1 kHz is good for SSB, and 10 Hz for CW.
5. Set the volume with the volume encoder.
6. Pick a filter that suits the mode: press FILT, or use the Audio panel.

5.1 Reducing noise and interference

The Baseband menu (Noise tab) has the UHSDR noise tools:

- **Noise reduction (NR)** reduces hiss on SSB and AM. Raise *NR strength* until the noise drops, and stop before the speech starts to sound watery.
- **Noise blanker (NB)** removes short impulse noise, such as ignition or electric fence clicks.
- **Auto notch** finds and removes steady carriers (tuning tones, heterodynes) on SSB and AM.
- **Manual notch** removes one tone at a frequency you set. **Peak filter** does the opposite: it boosts one tone, which helps to dig out a weak CW signal.

5.2 Receiving AM broadcast and weak AM stations

Switch on *AM as SyncAM (SAM)* on the RX tab of the Baseband menu. The synchronous detector locks to the carrier, which reduces fading distortion. *SAM fade leveler* makes the audio more even during selective fading.

6 Transmitting

Before you transmit

- Connect a suitable antenna or a 50 Ω dummy load. Transmitting without a load can damage the power amplifier.
- Start at low power. The default is 2 W.
- Check that you are inside your licence privileges.

6.1 Power level

Press **PWR** on the quick-access panel to step through 0.5, 1, 2, 5, 10, 15 and 20 W. Each band keeps its own setting. The TX status area on the main screen shows the selected level. BETA The power steps are not yet calibrated against a wattmeter.

6.2 Tune

Press **F5** to send a steady carrier, for example to adjust an antenna tuner. Press F5 again to stop. The **TUNE** button in the Power Amplifier menu does the same. F5 only switches the transmitter: it does not change the mode or the frequency. The front-panel TX LED lights while the radio transmits. WORKING

6.3 SSB

Press the microphone PTT to transmit, and release it to return to receive. BETA SSB transmit works. The microphone gain and speech compressor settings are still being tuned by ear. You can adjust them on the TX tab of the Baseband menu:

- **Mic gain**: set it so that normal speech gives full output without distortion.
- **Speech compressor**: raises the average power. The default is level 4. Higher levels give more “punch” but sound less natural.
- **SSB TX filter** and **TX bass/treble**: the shape of the transmitted audio.

6.4 CW

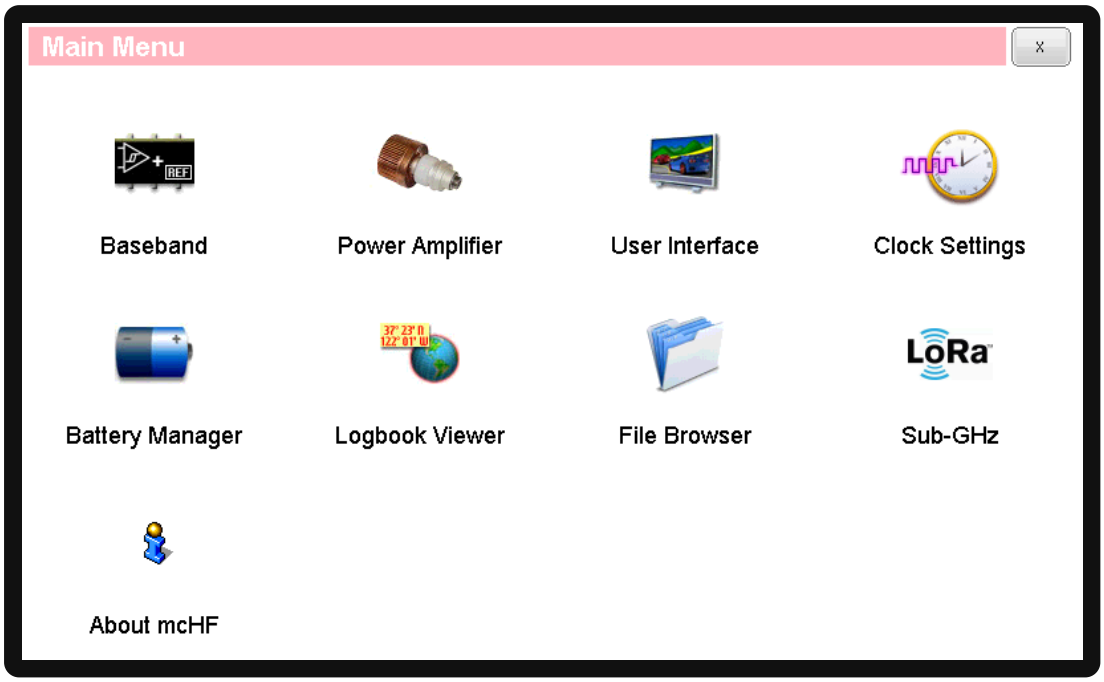
Connect a paddle to the key jack and select CW mode. The iambic keyer is built in; set it on the CW tab of the Baseband menu:

- **Keyer speed**: 5 to 48 WPM (default 12)
- **Keyer weight**: the dot/dash ratio
- **Sidetone/offset**: the sidetone pitch, which is also the receive offset (default 750 Hz)
- **Paddle reverse**: swaps the dot and dash paddles
- **TX to RX delay**: how long the radio stays in transmit after the last element

The volume encoder sets the sidetone volume while you send. WORKING

7 The menu

Press **F1** to open the menu, touch an icon to open a page, and press F1 again to return to the main screen.



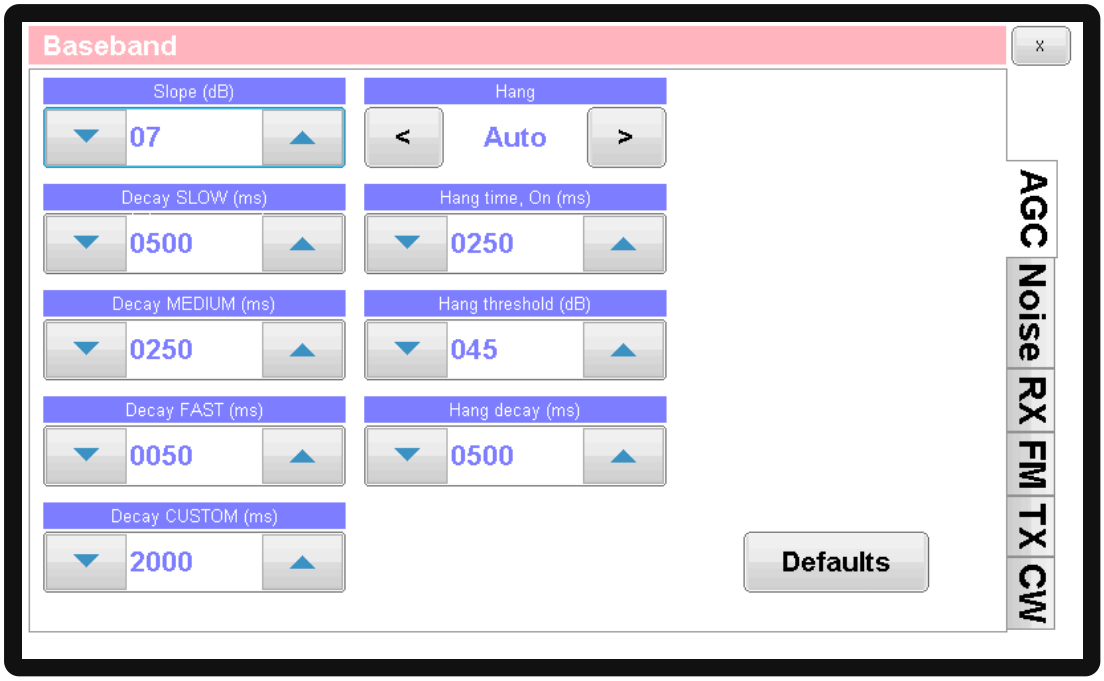
The main menu

Every menu page has an **X** button at the top right that returns to the main menu.

7.1 Baseband

All the receive and transmit DSP settings, on six tabs down the right edge: **AGC**, **Noise**, **RX**, **FM**, **TX** and **CW**. Numbers are changed with the ▼/▲ arrows, choices with < and >, and switches with the tick boxes. A change takes effect at once and is saved automatically. **Defaults** restores the factory values. BETA The menu and the keyer speed are confirmed on the radio. The other settings have not all been tested one by one.

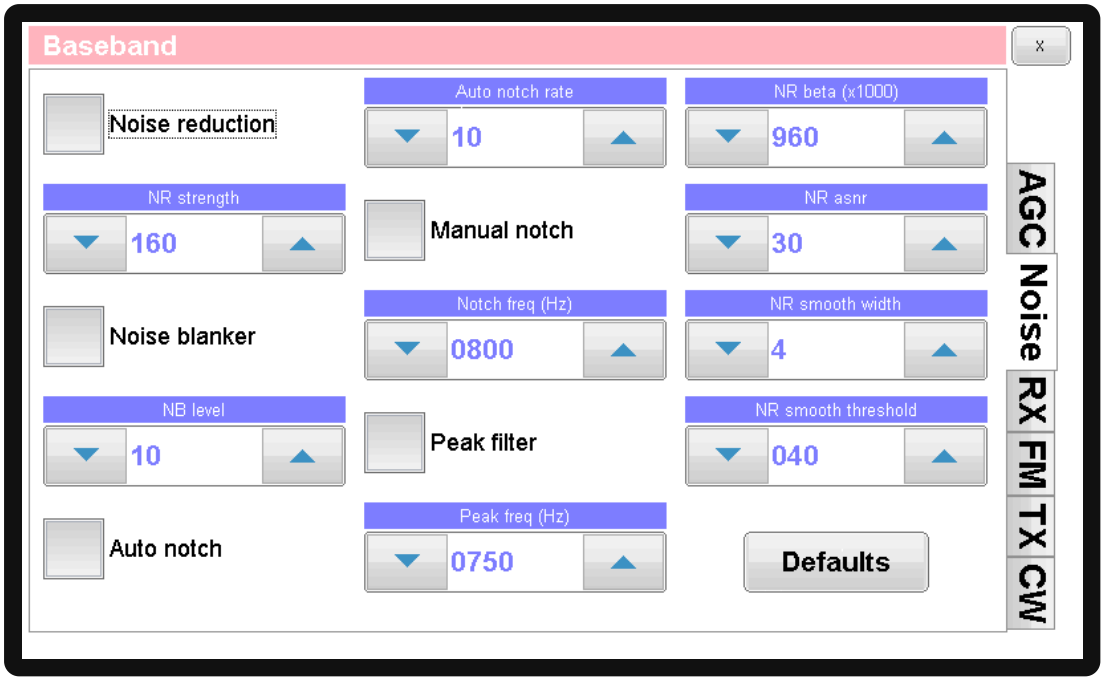
AGC tab



Baseband menu, AGC tab

Setting	Default	Range	Meaning
Slope (dB)	7	0 … 20	How much the audio still rises above the AGC threshold. 0 = flat
Decay SLOW (ms)	500	10 … 5000	Recovery time of the SLOW AGC mode
Decay MEDIUM (ms)	250	10 … 5000	Recovery time of the MEDIUM AGC mode
Decay FAST (ms)	50	10 … 5000	Recovery time of the FAST AGC mode
Decay CUSTOM (ms)	2000	10 … 5000	Recovery time of the CUSTOM AGC mode
Hang	Auto	Auto / Off / On	Auto uses the preset of each AGC mode
Hang time, On (ms)	250	10 … 5000	Gain is held this long after a signal stops (Hang = On)
Hang threshold (dB)	45	-20 … 120	Signals below this level do not start the hang
Hang decay (ms)	500	100 … 5000	Recovery time after the hang ends

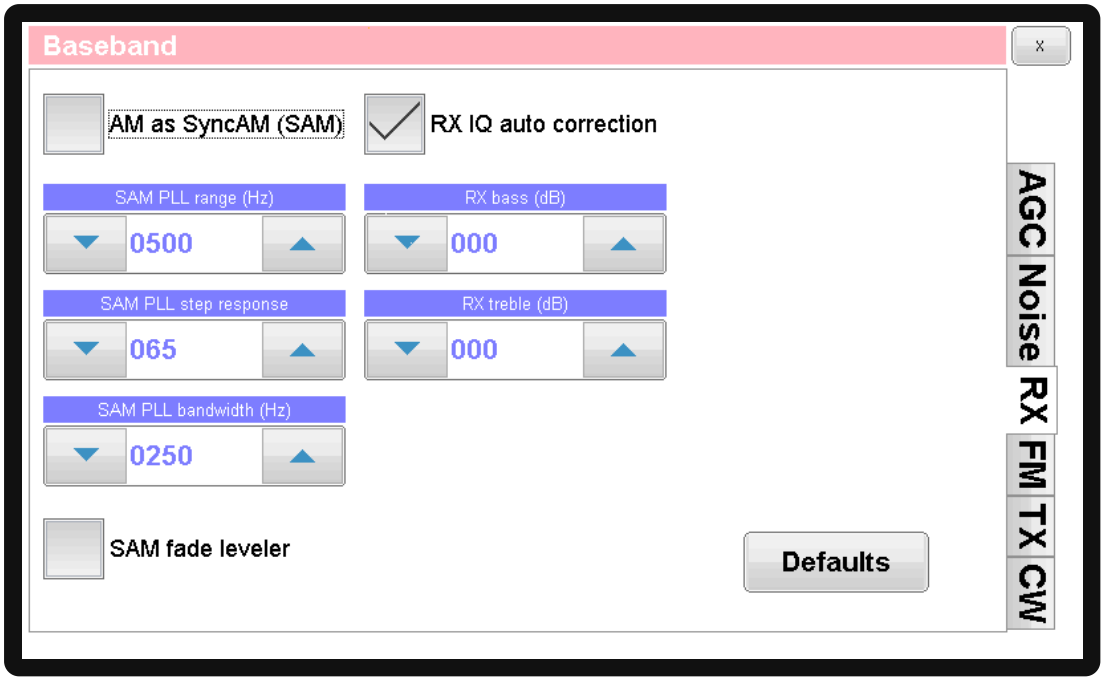
Noise tab



Baseband menu, Noise tab

Setting	Default	Range	Meaning
Noise reduction	Off	On / Off	Spectral noise reduction
NR strength	160	1 … 200	Higher removes more noise, but adds artefacts
Noise blanker	Off	On / Off	Removes short impulse noise
NB level	10	1 … 15	Blanker threshold, higher blanks more
Auto notch	Off	On / Off	Removes steady carriers automatically (SSB/AM)
Auto notch rate	10	0 … 40	How fast the auto notch adapts
Manual notch	Off	On / Off	Removes one tone at a fixed audio frequency
Notch freq (Hz)	800	200 … 5000	Audio frequency of the manual notch
Peak filter	Off	On / Off	Boosts one tone, for weak CW
Peak freq (Hz)	750	200 … 5000	Audio frequency of the peak filter
NR beta (x1000)	960	700 … 999	NR fine tuning: noise estimate smoothing
NR asnr	30	2 … 30	NR fine tuning: a-priori SNR
NR smooth width	4	1 … 5	NR fine tuning: spectral smoothing width
NR smooth threshold	40	10 … 100	NR fine tuning: smoothing threshold

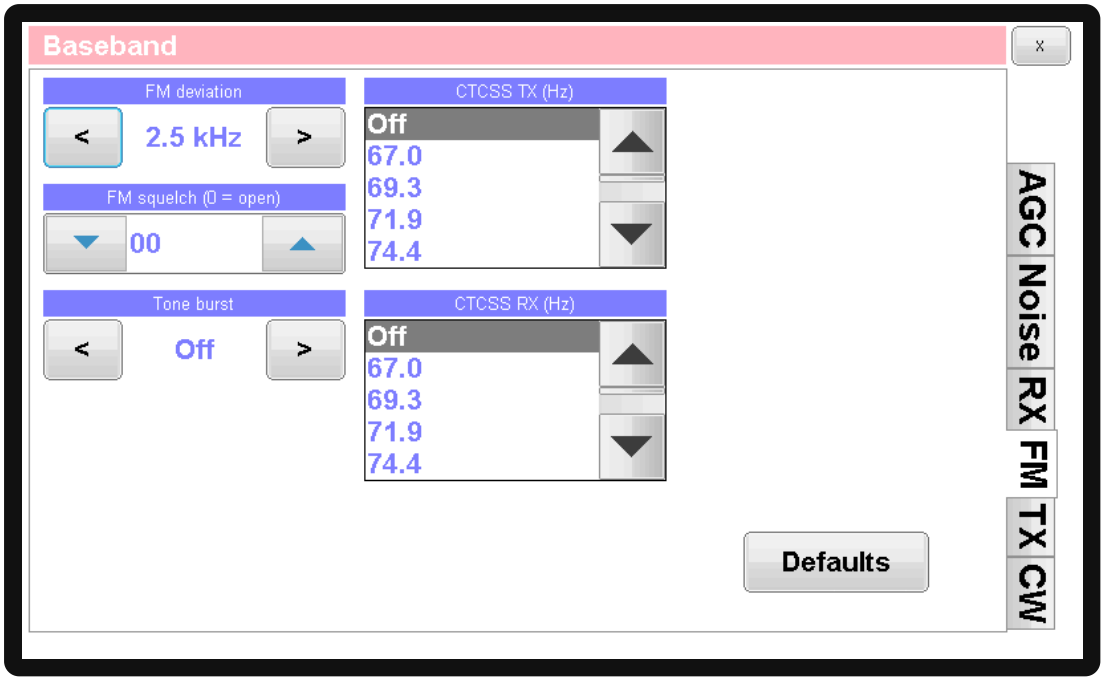
RX tab



Baseband menu, RX tab

Setting	Default	Range	Meaning
AM as SyncAM (SAM)	Off	On / Off	AM is received with the synchronous detector
SAM PLL range (Hz)	500	50 … 8000	How far the SAM PLL searches for the carrier
SAM PLL step response	65	1 … 100	PLL damping (step response)
SAM PLL bandwidth (Hz)	250	25 … 1000	PLL loop bandwidth
SAM fade leveler	Off	On / Off	Evens out audio level during selective fading
RX IQ auto correction	On	On / Off	Automatic image rejection. Leave on
RX bass (dB)	0	-20 … 20	Receive audio bass
RX treble (dB)	0	-20 … 20	Receive audio treble

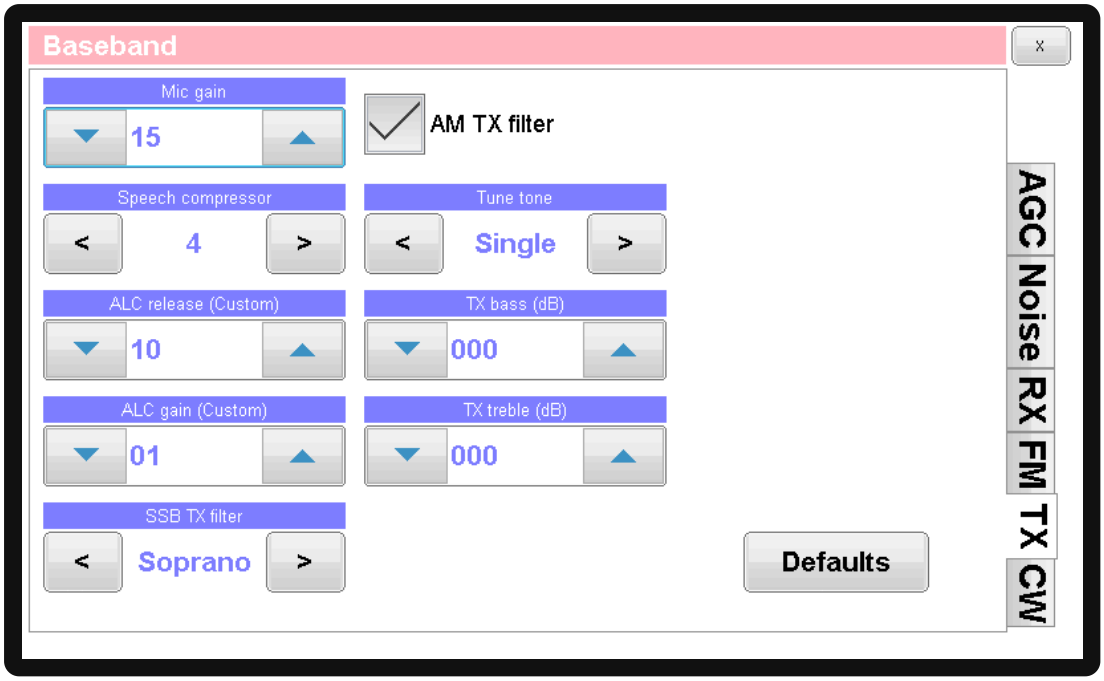
FM tab



Baseband menu, FM tab

Setting	Default	Range	Meaning
FM deviation	2.5 kHz	2.5 kHz / 5 kHz	FM transmit deviation
FM squelch (0 = open)	0	0 … 20	FM squelch level, 0 = always open
Tone burst	Off	Off / 1750 Hz / 2135 Hz	Repeater access tone burst
CTCSS TX (Hz)	Off	Off … 254.1	Sub-audible tone sent on transmit
CTCSS RX (Hz)	Off	Off … 254.1	Sub-audible tone needed to open the squelch

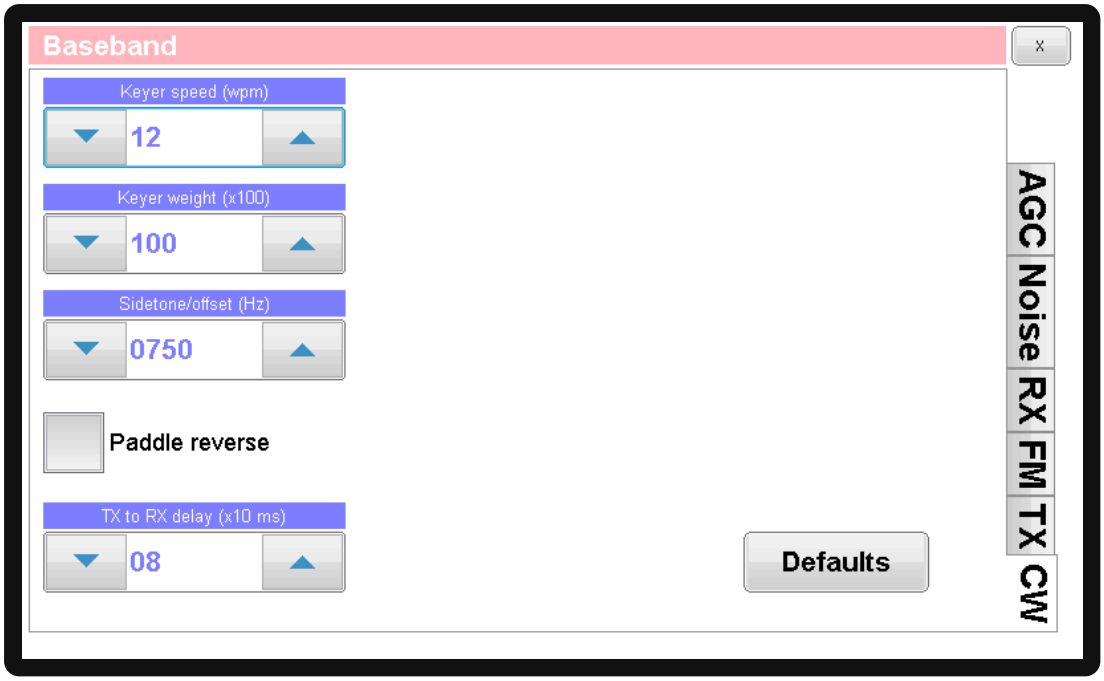
TX tab



Baseband menu, TX tab

Setting	Default	Range	Meaning
Mic gain	15	2 … 99	Microphone level
Speech compressor	4	Off … Custom	Speech compression; Custom uses the two ALC values
ALC release (Custom)	10	0 … 20	ALC release, Custom compressor only
ALC gain (Custom)	1	1 … 25	ALC gain before compression, Custom only
SSB TX filter	Soprano	Soprano / Tenor / Bass	Transmit audio shape: Soprano is brightest
AM TX filter	On	On / Off	Band-limits the AM transmit audio
Tune tone	Single	Single / Two tone	Carrier used by TUNE
TX bass (dB)	0	-20 … 5	Transmit audio bass
TX treble (dB)	0	-20 … 5	Transmit audio treble

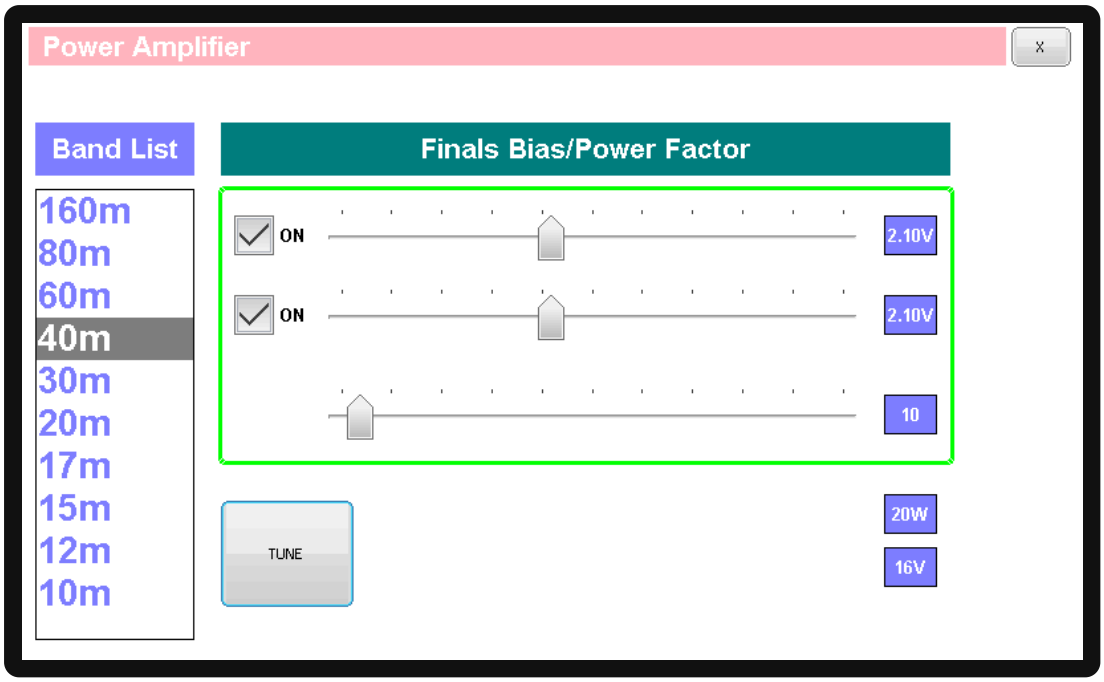
CW tab



Baseband menu, CW tab

Setting	Default	Range	Meaning
Keyer speed (wpm)	12	5 … 48	Iambic keyer speed
Keyer weight (x100)	100	50 … 150	Dot/dash weighting, 100 = standard
Sidetone/offset (Hz)	750	400 … 1000	Sidetone pitch, also the CW receive offset
Paddle reverse	Off	On / Off	Swap dot and dash paddles
TX to RX delay (x10 ms)	8	0 … 50	Time in transmit after the last element

7.2 Power Amplifier

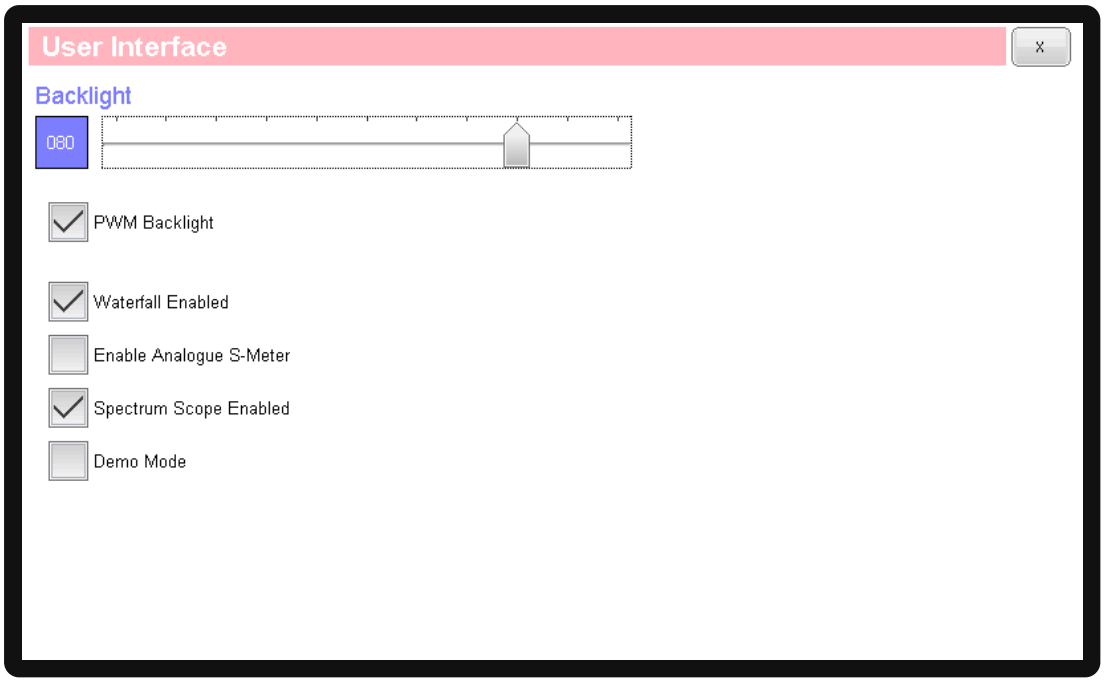


Power Amplifier menu

- **Band List**: selects the band whose settings are shown
- **Finals Bias/Power Factor**: the bias voltage of each of the two final transistors (each with its own ON switch), and the power factor of the band, which scales the output so that the power steps come out right
- **TUNE**: sends a carrier, like F5. Use it to check the settings with a wattmeter and a dummy load.

> The bias and power factor are calibration settings. A wrong bias can destroy the final transistors. Change them only if you know what you are doing. BETA

7.3 User Interface

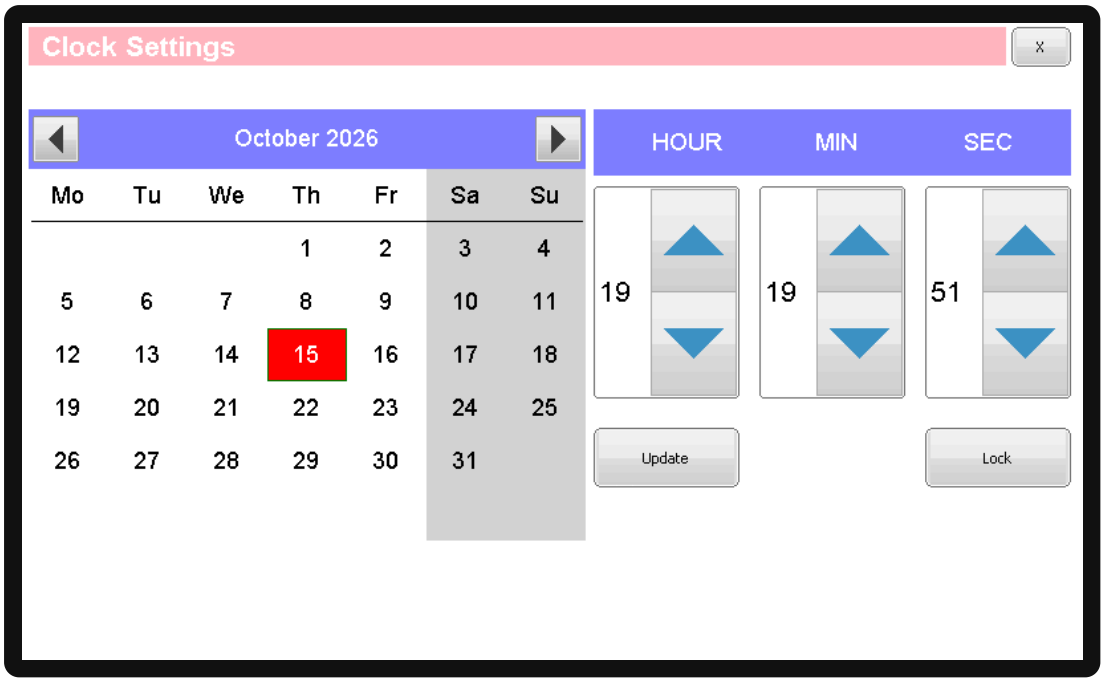


User Interface menu

- **Backlight**: display brightness
- **PWM Backlight**: dims the backlight by PWM. Leave on.
- **Waterfall Enabled** and **Spectrum Scope Enabled**: show or hide them on the main screen
- **Enable Analogue S-Meter**: the needle meter when ticked, the bar meter when not
- **Demo Mode**: sweeps the frequency by itself, for exhibitions
- **Quick launch keys**: the app that F2, F3 and F4 open (FT8, MeshCore, MarsChat, WSPR or None). Picking an app that is already on another key swaps the two keys. See Quick launch keys.

WORKING

7.4 Clock Settings

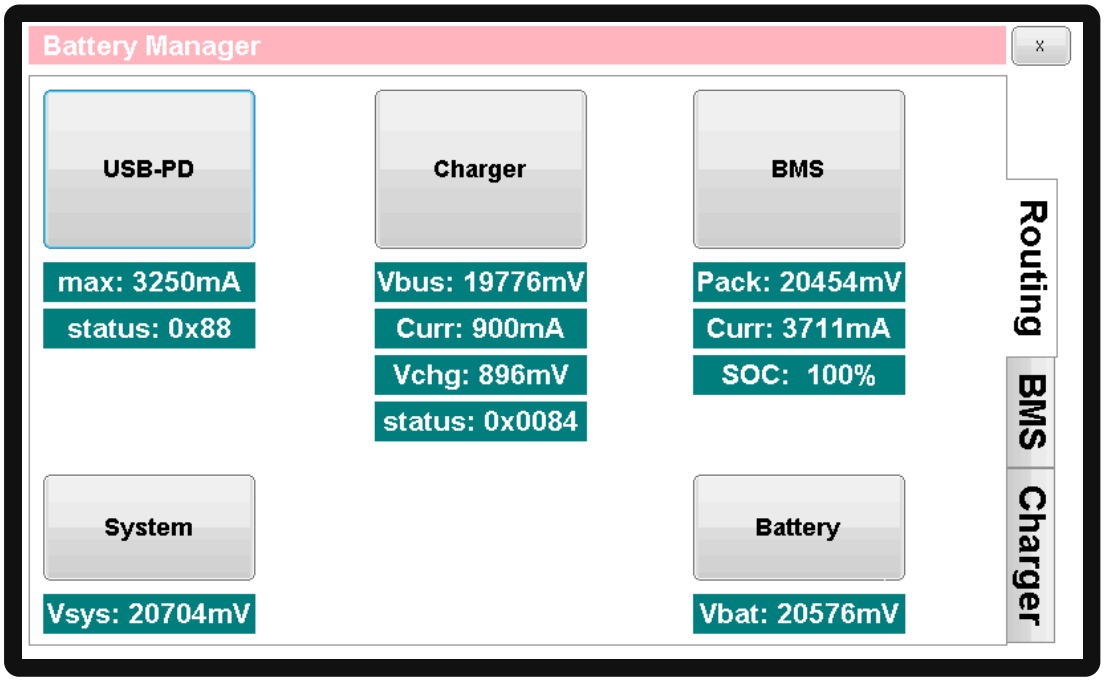


Clock Settings

Touch the date in the calendar and set the hour, minute and second with the arrows, then touch **Update** to set the clock. On radios with GPS the clock sets itself (see Time and GPS), so you only need this page without GPS. Keep the clock on **UTC**: FT8, WSPR and MarsChat all work in UTC time slots. WORKING

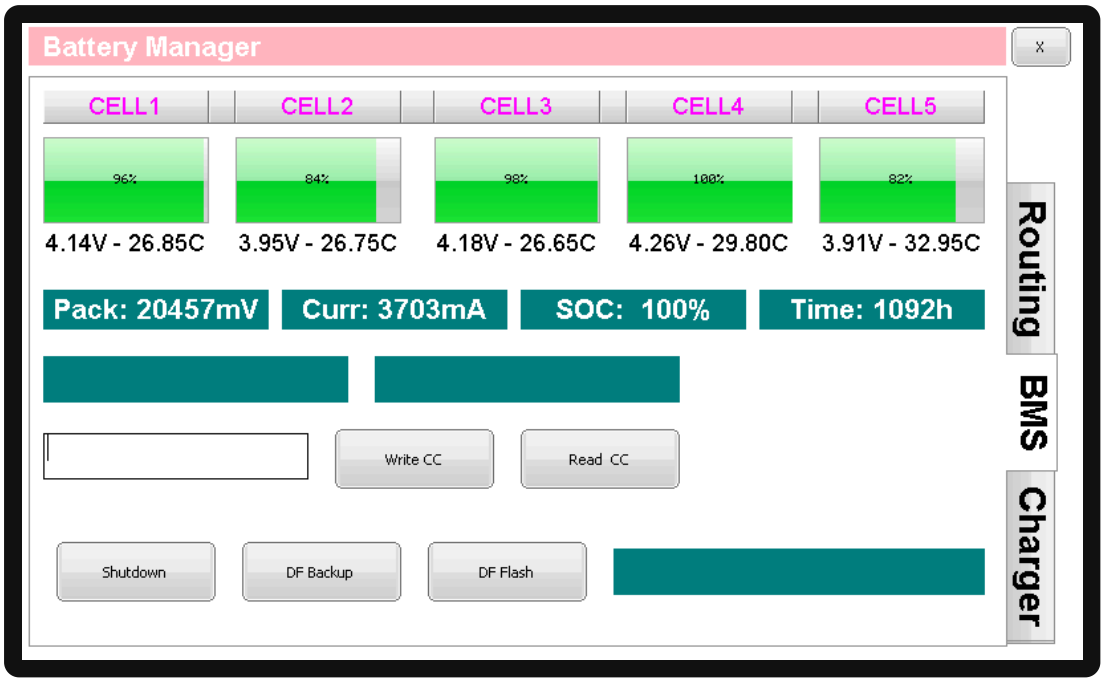
7.5 Battery Manager

The battery pack, the charger and the power paths, on three tabs: **Routing**, **BMS** and **Charger**. WORKING



Battery Manager, Routing tab

The **Routing** tab shows where the power comes from and goes to: the USB-PD contract (maximum current), the charger (input voltage and current), the battery gauge (pack voltage, current and charge), and the system and battery rail voltages.



Battery Manager, BMS tab

The **BMS** tab shows each of the five cells (charge, voltage and temperature), the pack voltage and current, the state of charge (SOC) and the operating time. It also has:

- **Shutdown**: switches the radio off
- **Read CC** / **Write CC**: current-sense calibration of the battery gauge
- **DF Backup** / **DF Flash**: saves the battery gauge configuration to the SD card, or loads it back

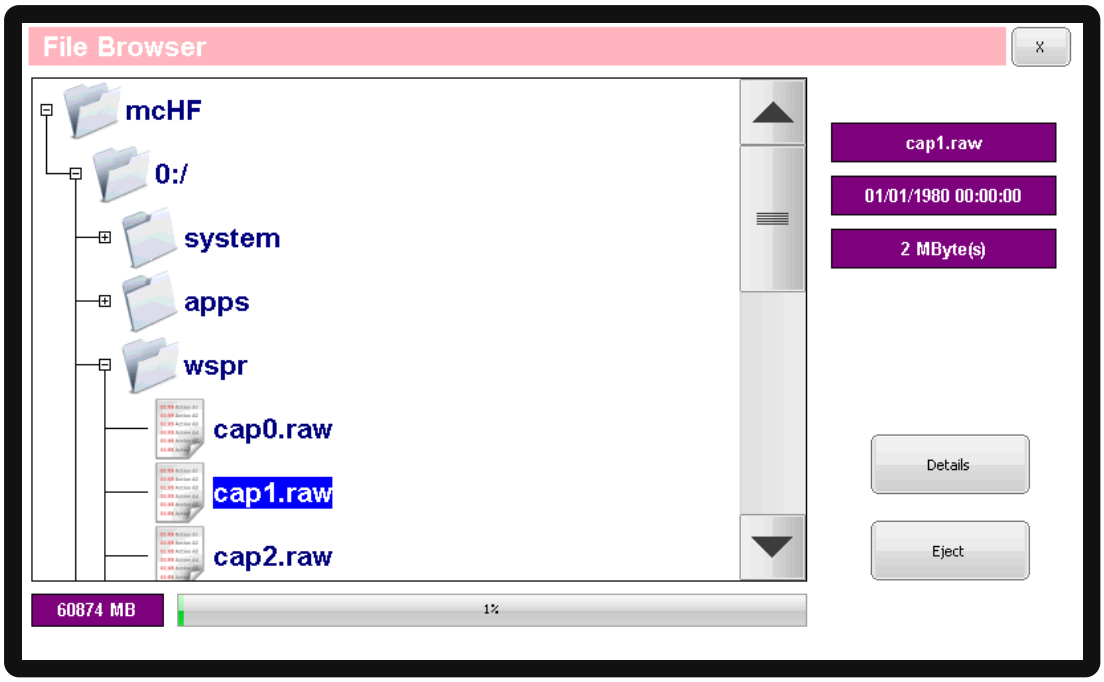
> Read CC, Write CC, DF Backup and DF Flash are service functions. Do not use them unless you are asked to: a wrong gauge configuration can stop the battery from charging or discharging.

The **Charger** tab shows the charger settings and status.

7.6 Logbook Viewer

PLANNED The logbook viewer layout exists, but it shows sample data. FT8 contacts are logged to the SD card (see FT8), but they do not show here yet.

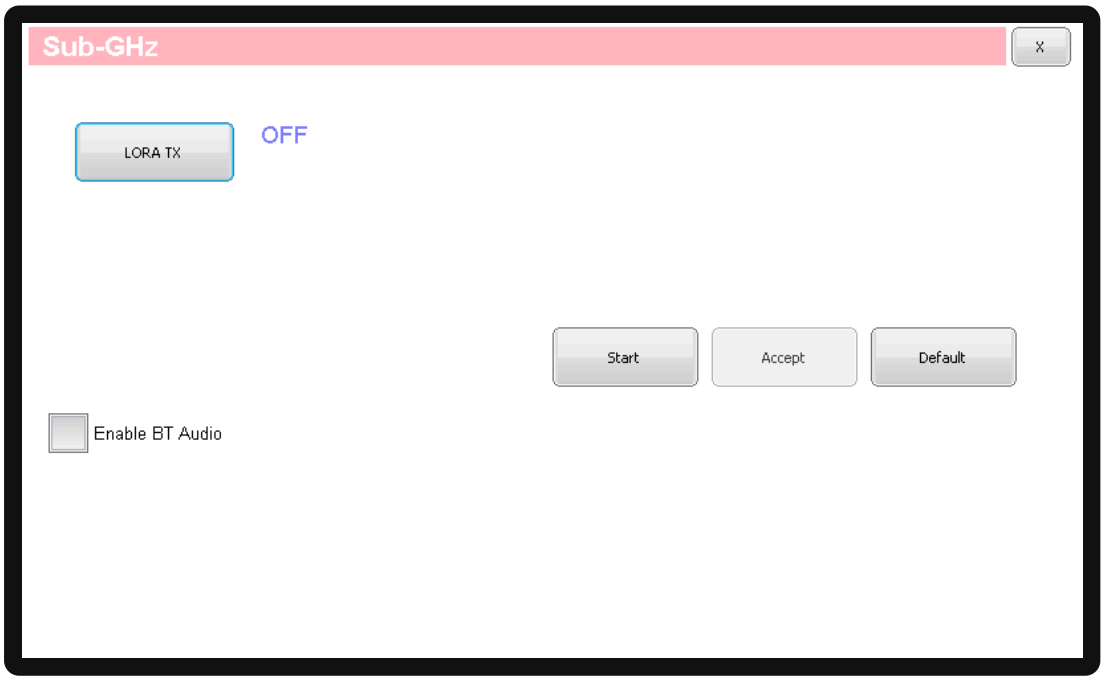
7.7 File Browser



File Browser

Browses the SD card as a folder tree. Touching a file shows its name, date and size on the right. **Details** opens the file properties, where the file can be deleted, and **Eject** prepares the card for removal. The bar at the bottom shows the card size and how full it is. BETA

7.8 Sub-GHz



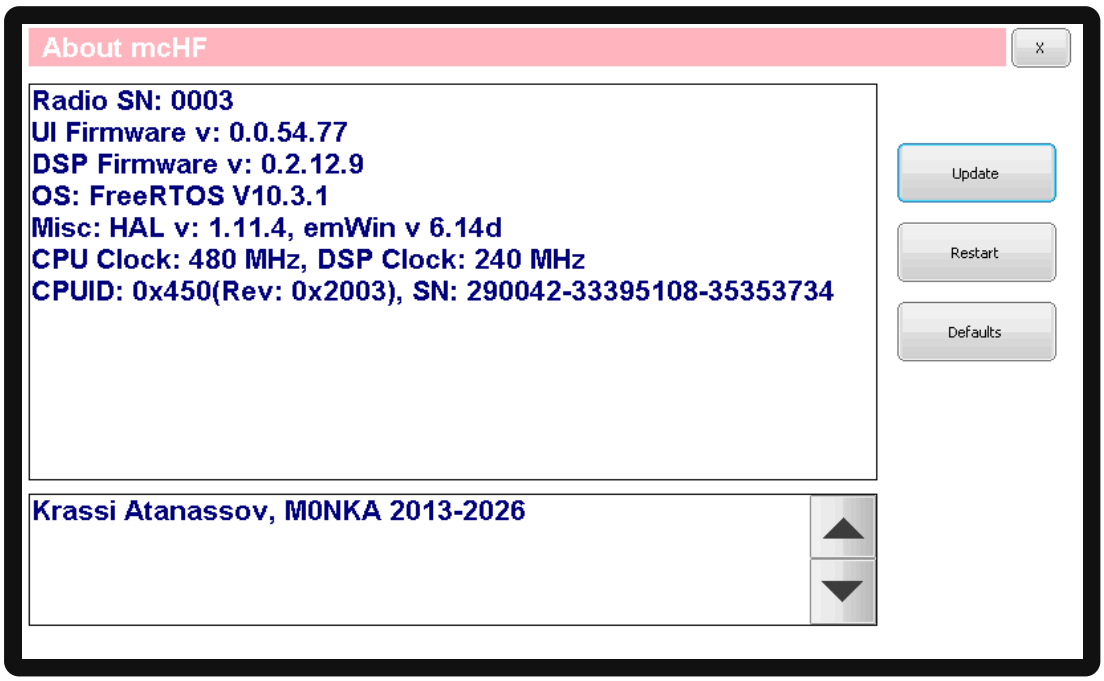
Sub-GHz menu

- **LORA TX**: switches the LoRa transmitter on and off

- **Start / Accept / Default**: will measure the clock crystal error against the GPS and trim the clock. PLANNED
- **Enable BT Audio**: Bluetooth audio PLANNED

BETA

7.9 About mcHF



About mcHF

The radio serial number, the firmware versions of the UI (M7) and DSP (M4) processors, the operating system and library versions, and the processor details. Quote these lines when you report a problem.

- **Update**: restarts the radio into the bootloader, for a firmware update
- **Restart**: restarts the radio
- **Defaults**: sets all the baseband (DSP) parameters back to their default values

WORKING

8 FT8

FT8 is the popular weak-signal digital mode. The mcHF decodes and transmits FT8 by itself, without a computer. Press **F2** (the default quick launch key) to open the FT8 screen and F2 again to leave it. The FT8 screen takes control of the HF radio while it is open. BETA Complete contacts work between two mcHF radios. On-air testing has just started.



The FT8 screen

8.1 Before you start

- **Set your callsign** in the `[station]` section of the settings file. FT8 and WSPR share it.
- **The clock must be correct to within about one second.** FT8 uses 15-second slots aligned to UTC. With GPS, wait until *lock* appears next to the time on the main screen. Without GPS, set the clock carefully in Clock Settings. If the screen shows many candidates but no decodes, check the clock first.
- Touch **BAND** to step through the bands. The radio tunes to the standard FT8 frequency of each band.

8.2 Screen layout

- The **top line** shows the band, the dial frequency and the UTC time.
- The **status bar** shows the transmitter state (RX ONLY, or the QSO state), the receiver state, the number of decodes in the last slot (DEC), the measured clock offset in seconds (CLK) and the time to the next slot.
- The **waterfall** shows the audio passband, 0 to 3000 Hz. Every FT8 signal is a short vertical trace.
- **BAND ACTIVITY** lists every station decoded in the last slots, with UTC time, signal report (DB), time offset (DT), audio frequency and message. CQ calls are shown in orange.
- **RX FREQUENCY** shows only the traffic to and from you and your current partner.

- The **TX strip** under the lists shows the state of the contact (CALLING, SENT REPORT, RR73 and so on) and the next transmit slot (EVEN or ODD).

8.3 Making a contact

1. Touch **ENABLE TX** to arm the transmitter (*TX ARMED*). While it is disarmed the radio only receives.
2. To call CQ, touch **CALL CQ**. To answer someone, touch **ANSWER**. It picks the next CQ in the band activity list and replies to it. Touch it again to move on to the next CQ.
3. The radio runs the rest of the contact by itself: report, R+report, RR73 and 73.
4. Completed contacts are written to `ft8/qso.txt` on the SD card, and every decode goes into `ft8/decodes.txt`.
5. Touch **HALT TX** to stop transmitting at any time.

The **LOG** button is reserved for the logbook. PLANNED

> **Current limitations**
>
> - The callsign is set in the settings file on a PC. An edit box in the menu is planned.
> - Signal reports are estimated from the decoder score, not yet measured in dB.
> - Touching a station in the list to answer it, and ADIF logging, are planned.

9 MarsChat

MarsChat is a keyboard-to-keyboard text mode for very weak signals. It is designed for planet-wide contact with no infrastructure at all. It uses the WSPR signal format (two-minute slots, very narrow bandwidth), so text works at signal levels far below what you can hear, though it is slow: a few characters per two-minute slot. Delivery is confirmed: each message is acknowledged, and the radio sends it again if it was not.

Press **F4** (the default quick launch key) to open MarsChat and F4 again to leave it. BETA Tested between two mcHF radios on the bench.



The MarsChat screen

9.1 Starting a session

Both stations transmit in turn, in alternating two-minute slots. So the two stations must take different roles:

- The station that starts the chat is the **CALLER**. It just writes a message and touches SEND. Sending starts the session in the caller role.
- The station that answers touches **PEER** first, then waits for the caller's messages.

If both stations choose the same role, they transmit in the same slots and never hear each other. Touching the active role button again ends the session. The bar at the top shows the role and what the current slot is for (OUR SLOT or PEER SLOT), and the time to the next receive.

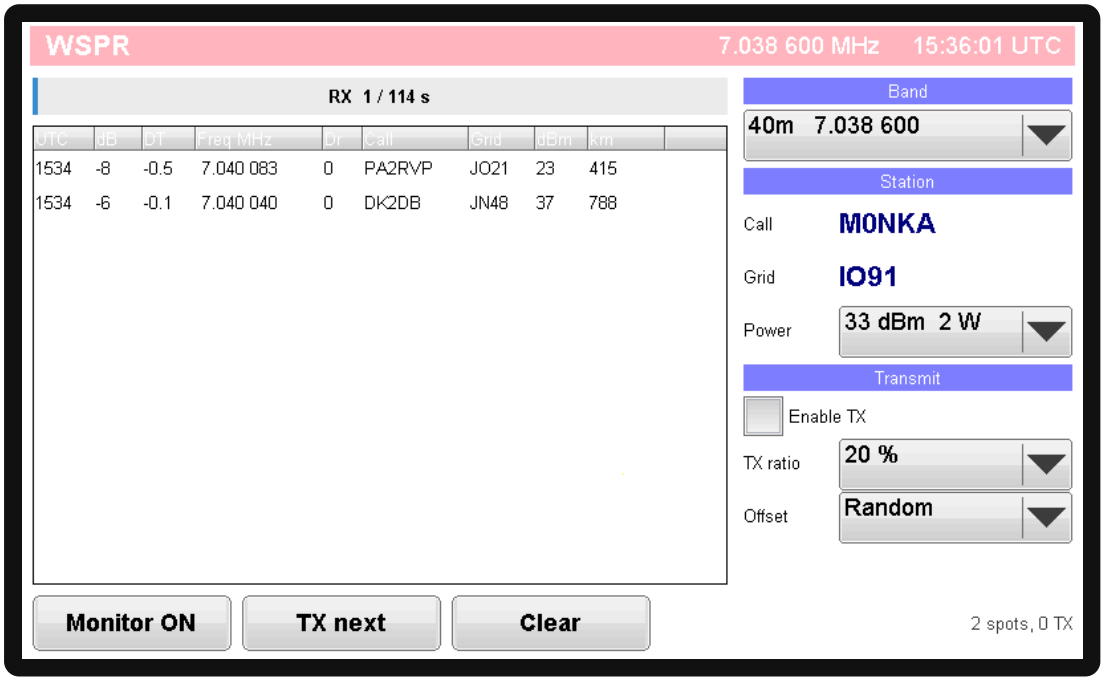
9.2 Screen layout

- Touch **TYPE** to open the on-screen keyboard, write your message, then touch **SEND**. The message goes into the **QUEUE** and is sent in your next slot.
- The **SLOT TIMELINE** shows the 120-second slots: your slot (OUR SLOT) and your partner's slot (PEER SLOT) alternate.
- The **message area** on the left shows the chat. The panel on the right shows the session counters: the transmit sequence number (TX SEQ), the last acknowledgement sent (ACK SENT), the number of tries, the queue length, and the signal report (LAST SNR) and time offset (LAST DT) of the last received frame.
- The **TX** line shows how much text is waiting and how long it will take to send.
- **SIGNAL** shows the strength of the frame being received.
- If the other station stops answering, the screen shows *PEER SILENT – SEND TO RETRY*.

As with FT8, the clock must be correct. MarsChat uses the WSPR band segments. The full protocol is described in `doc/MarsChat_Protocol_Specification_v1.md`.

10 WSPR

WSPR (Weak Signal Propagation Reporter) is a beacon mode. Stations send their callsign, locator and power in two-minute slots, and every station that decodes them learns how the band is open. The mcHF receives and decodes WSPR by itself and can also transmit its own beacon. WSPR has no key by default: set one of F2–F4 to **WSPR** in the User Interface menu, then press that key to open the screen and again to leave it. The WSPR screen takes control of the HF radio while it is open. BETA Receive and decoding work on the air. Transmitting is not tested yet.



The WSPR screen with two spots on 40 m

10.1 Before you start

- Set your callsign and locator in the `[station]` section of the settings file. Without them the radio only receives, and the status line says why.
- The clock must be correct to within about one second, as for FT8 (see Time and GPS).

10.2 Screen layout

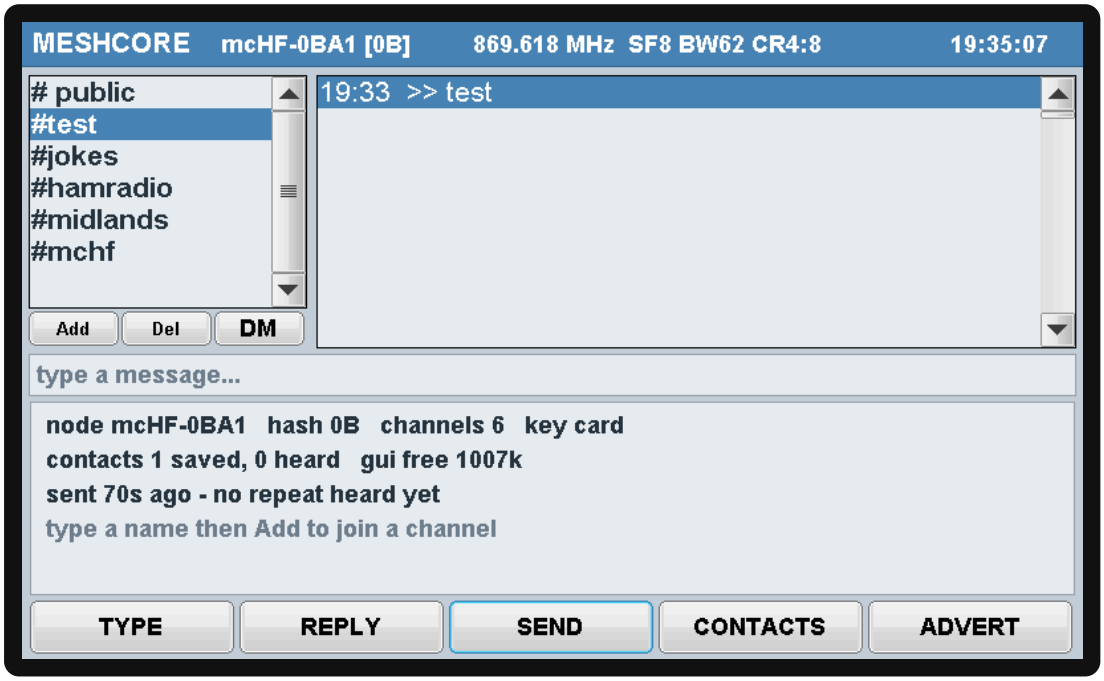
- The **title bar** shows the dial frequency and the UTC time.
- The **cycle bar** shows where the radio is in the two-minute cycle: waiting for the next even minute, receiving (RX, 114 seconds), decoding, or transmitting (TX, in red).
- The **spot list** shows every station decoded, newest first: UTC time, signal-to-noise ratio (dB), time offset (DT), frequency, drift (Dr), callsign, locator, reported power (dBm) and distance (km). Touch a column header to sort by it.
- **Band** selects the band. The radio tunes to the standard WSPR dial frequency of that band.
- **Station** shows the callsign and locator from the settings file, in red when they are missing.

- **Power** is the power the beacon transmits and reports, in dBm and watts.
- **Transmit**: **Enable TX** allows the beacon. **TX ratio** is the share of cycles that transmit (20 % is the usual WSPR setting), chosen at random so that two stations don't keep colliding. **Offset** is the transmit frequency inside the 200 Hz WSPR window (Random by default).
- **Monitor** switches receiving on and off. **TX next** transmits in the next cycle instead of waiting for the random choice. **Clear** empties the spot list.
- The **status line** at the bottom right shows the number of spots and transmissions, or why the radio may not transmit.

> With **Enable TX** ticked, the radio transmits on its own in some of the cycles. Connect an antenna or a dummy load first.

11 MeshCore (LoRa)

MeshCore is an off-grid text messaging network that runs on small LoRa radios. Messages hop from node to node through repeaters. The mcHF talks to the MeshCore network through its built-in LoRa radio, so you can chat with MeshCore users and nodes nearby. Press **F3** (the default quick launch key) to open the MeshCore screen and F3 again to leave it. WORKING Channel messages work in both directions on the live network.



The MeshCore screen

- The **top line** shows your node name, the LoRa frequency and the radio settings (spreading factor, bandwidth, coding rate) and the UTC time.
- The **channel list** on the left has the public channel and hashtag channels (for example `#test`). Touch a channel to show its messages on the right. To join a channel, type its name and touch **Add**. **Del** leaves the selected channel.
- Touch **TYPE** to open the keyboard, write the message and touch **SEND**. **REPLY** answers the selected message.
- The **status area** shows your node and its hash, the number of channels and contacts, and whether your last message was repeated by the mesh.
- **CONTACTS** lists the nodes the radio has heard: chat nodes, repeaters, room servers and sensors. **ADD** saves a contact, and **FORGET** removes it.
- **ADVERT** announces your node to the network, so others can find you.
- **DM** sends a direct message to a saved contact. BETA

Your node identity (its key pair), contacts and channels are kept in the `meshcore/` folder on the SD card. MeshCore keeps receiving in the background while you use the HF radio, and new messages appear as notifications on the main screen.

12 Time and GPS

Digital modes need an accurate UTC clock. Rev B radios have a GPS receiver that sets the clock in two stages:

1. **Time lock.** As soon as the receiver sees satellites, usually within about 20 seconds even indoors near a window, it sets the hours, minutes and seconds. The clock panel on the main screen then shows *lock* next to the time. This is all FT8, WSPR and MarsChat need.
2. **Full lock.** With a good view of the sky, after a few minutes, the date and position are added.

While GPS time is valid, the clock is corrected every 30 seconds. Without GPS, the battery-backed clock keeps time with a trimmed crystal, but check it from time to time. WORKING

13 Battery and charging

The internal 80 Wh pack (five lithium cells in series) is managed by a smart battery gauge, which monitors the cell voltages, temperatures and charge. The pack charges from a USB-C power-delivery charger. The battery icon on the main screen shows the charge level, and the Battery Manager menu shows the details.
WORKING

- Use a good-quality USB-C PD charger. The Charger tab of the Battery Manager shows the contract the charger offers.
- Transmitting at high power draws far more current than receiving, so a full battery gives many hours of listening but less transmit time.

13.1 Removing and replacing the cells

The pack is made of five 21700 cells in a compartment at the back of the radio. Before you open the case for any reason, take the cells out first, as follows:

1. Switch the radio off with the POWER key.
2. Unplug the USB cable and any other cables connected to the radio.
3. Open the battery lid at the back. With a small screwdriver, take the cells out in the order 5, 4, 3, 2, 1 (right to left).

When the radio is back together, put the cells in the order 1, 2, 3, 4, 5 (left to right). Check the polarity of every cell before you insert it.

> **Important**
>
> - Lever the cells out with the screwdriver at the **negative** terminal only.
> - The mcHF supports only Samsung INR21700-40T cells, or an equivalent.

14 The SD card

Use a microSD card formatted FAT32. The radio creates these folders as it needs them:

Folder	Contents
`screens/`	Screenshots, `scr_0001.bmp` …
`ft8/`	`qso.txt` (completed contacts), `decodes.txt` (all decodes)
`meshcore/`	Node identity, contacts and channels
`system/`	`settings.ini`, the settings file
`wspr/`	WSPR / MarsChat working files: recordings of the last eight receive cycles, `cap0.raw` to `cap7.raw`, reused in turn
`bms/`	Battery gauge configuration backups

> Keep a copy of the `meshcore/` folder. It holds your node's private key, which is your identity on the MeshCore network.

14.1 The settings file

When you switch the radio off, it writes all its settings to `system/settings.ini` on the SD card: the band, brightness, S-meter style, AGC and RF gain, the quick launch keys, the frequencies, mode, filter, step and power of every band, and the baseband (DSP) parameters. At power on the radio reads the file back. So the settings move with the card, and you can keep a copy of them on a PC. **WORKING**

The file is plain text and fine to edit on a PC. A changed file is loaded at the next power on. Unknown keys are ignored, and values out of range are corrected or skipped. A few settings exist only in the file, for now:

```
[station]
call=M0NKA
grid=IO91
```

`call` and `grid` are your callsign and four-character locator, used by FT8 and WSPR. Leave them empty and the digital modes only receive.

> Edit the file with the radio switched off, then put the card back before you switch on. If you edit it while the radio is on, the radio overwrites your changes when it is switched off.

15 Screenshots

Hold F1 for about half a second to save the screen to the SD card. The display freezes for a moment while the picture is saved. On the main screen, a notification then shows the file name. The pictures are 800 × 480 pixel BMP files in the `screens/` folder, numbered `scr_0001.bmp`, `scr_0002.bmp` and so on. This works on the main screen, in the menu and on the FT8, MarsChat, MeshCore and WSPR screens. BETA

The screenshots in this manual were made this way. They are also the best way to report a display problem.

16 Feature status summary

Feature	Status	Notes
SSB / CW / AM receive	WORKING	
Synchronous AM	BETA	Baseband menu, RX tab
FM	PLANNED	In the DSP and settings, but it cannot be selected from the front panel yet
Spectrum scope and waterfall	WORKING	
S-meter	BETA	Not calibrated: it reads high
Noise reduction, blanker, notch	BETA	From UHSDR, not all tested on v0.9
AGC modes and RF gain	BETA	
CW transmit and keyer	WORKING	
SSB transmit	BETA	Audio still being tuned
Power level steps	BETA	Not calibrated
FT8	BETA	Callsign from the settings file
MarsChat	BETA	
WSPR	BETA	Receive and decoding work. Transmit not tested yet.
Quick launch keys (F2–F4)	WORKING	Set in the User Interface menu
Settings file on the SD card	WORKING	`system/settings.ini`
MeshCore	WORKING	Direct messages are beta
GPS time	WORKING	Rev B boards
Battery management	WORKING	
Screenshots	BETA	
Logbook	PLANNED	
Bluetooth audio	PLANNED	
CAT control / USB audio	PLANNED	

17 Credits

The mcHF builds on the work of many radio amateurs and open source projects:

- **UHSDR** (DF8OE, DB4PLE, DD4WH and the UHSDR team), the DSP firmware that runs on the M4 core, with WDSP-derived AGC and noise reduction
- **ft8_lib** by Karlis Goba, YL3JG, the FT8 encoder and decoder
- **MeshCore**, the LoRa mesh protocol
- **FreeRTOS**, **FatFS** (ChaN), **STM32Cube** and **emWin** (SEGGER / ST)

Discussion and support: https://groups.io/g/mcHF/

18 Revision history

Edition	Date	Firmware (M7 / M4)	Changes
1.2	31 October 2026	v0.54.93 / v2.12.9	WSPR app; quick launch keys; settings file on the SD card; shake the dial to change the step; FT8 callsign from the settings file
1.1	18 October 2026	v0.54.77 / v2.12.9	Front panel photo; removing and replacing the battery cells; revision history
1.0	15 October 2026	v0.54.77 / v2.12.9	First edition